

Problem 8.27. Recall that a directed graph is strongly connected if every two nodes are connected by a directed path in each direction. Let

$$STRONGLY-CONNECTED = \{\langle G \rangle \mid G \text{ is a strongly connected graph}\}.$$

Show that *STRONGLY-CONNECTED* is NL-complete.

Proof. To show that *STRONGLY-CONNECTED* is NL-complete, we must show that it is in NL and that all NL problems are log space reducible to it. To show *STRONGLY-CONNECTED* $\in$ NL, we give an algorithm *S* that decides *STRONGLY-CONNECTED* in non-deterministic log space.

S = “On input $\langle G \rangle$, where *G* is a directed graph:

1. Non-deterministically select every distinct pair of nodes *s* and *t* in *G*:
2. Run the algorithm *M* for $\overline{PATH}$ given in Theorem 8.27 on $\langle G, s, t \rangle$.
3. *Accept*, if *M* accepts. *Reject*, if *M* rejects.”

To prove that *STRONGLY-CONNECTED* is NL-hard, we give a log space reduction from *PATH* to *STRONGLY-CONNECTED*. This reduction converts a directed graph *G* and two nodes *s* and *t* to a directed graph *G'* so that *G* has directed path from *s* to *t*, iff *G'* is strongly connected.

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